

Apriori is one of the easiest algorithms once you see it with a shopping basket example.

## Example: Supermarket Transactions

Suppose we have 5 customer baskets:

| Transaction | Items               |
|-------------|---------------------|
| T1          | Milk, Bread, Butter |
| T2          | Milk, Bread         |
| T3          | Milk, Diapers, Beer |
| T4          | Bread, Butter       |
| T5          | Milk, Bread, Butter |

Let's set:

- **Minimum Support = 40%**
- We have 5 transactions
- Therefore an itemset must appear in at least **2 transactions** (40% of 5 = 2)

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## Step 1: Count Single Items

Count how many times each item appears.

| Item    | Count |
|---------|-------|
| Milk    | 4     |
| Bread   | 4     |
| Butter  | 3     |
| Diapers | 1     |
| Beer    | 1     |

Keep only frequent items:

| Item   | Count |
|--------|-------|
| Milk   | 4     |
| Bread  | 4     |
| Butter | 3     |

Remove:

✗ Diapers (1)

✗ Beer (1)

These don't meet support.

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## Step 2: Create Candidate Pairs

Using survivors:

- {Milk, Bread}
- {Milk, Butter}
- {Bread, Butter}

Count them.

| Pair          | Count |
|---------------|-------|
| Milk, Bread   | 3     |
| Milk, Butter  | 2     |
| Bread, Butter | 3     |

All meet minimum support.

So all survive.

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## Step 3: Generate Triple

Since all pairs survived:

Candidate:

{Milk, Bread, Butter}

Count occurrences:

Appears in:

- T1
- T5

Count = 2

Support =  $2/5 = 40\%$

 Frequent

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## Step 4: Apriori Pruning Trick

This is the MOST IMPORTANT part.

Suppose we wanted to test:

{Milk, Beer, Bread}

Before counting, Apriori asks:

"What are its subsets?"

Subsets:

- {Milk, Beer}
- {Milk, Bread}
- {Beer, Bread}

Earlier, Beer was infrequent.

Therefore:

{Milk, Beer} is infrequent.

Since one subset is infrequent:

✗ {Milk, Beer, Bread} can NEVER be frequent.

No need to count it.

This is Apriori's magic.

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## Why?

Imagine:

{Milk, Beer} appears only once.

Can:

{Milk, Beer, Bread}

appear 3 times?

Impossible.

Because every occurrence of:

{Milk, Beer, Bread}

must also contain:

{Milk, Beer}

So the larger set can never occur more often than its subsets.

This idea is called:

"If an itemset is frequent, all its subsets must also be frequent."

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## Step 5: No Larger Itemsets

We found:

- Frequent 1-itemsets
- Frequent 2-itemsets
- Frequent 3-itemsets

No 4-itemsets possible.

Stop.

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## Step 6: Generate Association Rules

From:

{Milk, Bread, Butter}

we can create rules.

### Rule 1

Milk  $\rightarrow$  Bread

Support:

Milk & Bread = 3

Confidence:

$$\frac{3}{4} = 75\%$$

Interpretation:

75% of customers who buy Milk also buy Bread.

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## Rule 2

Butter → Bread

Butter appears 3 times.

Butter & Bread appears 3 times.

$$\frac{3}{3} = 100\%$$

Interpretation:

Everyone who bought Butter also bought Bread.

This is a very strong rule.

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# Visual Understanding

Think of Apriori like a knockout tournament.

## Round 1

Single Items

|         |   |
|---------|---|
| Milk    | ✓ |
| Bread   | ✓ |
| Butter  | ✓ |
| Beer    | ✗ |
| Diapers | ✗ |

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## Round 2

Create pairs only from survivors

Milk-Bread   
Milk-Butter   
Bread-Butter 

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### Round 3

Create triples

Milk-Bread-Butter 

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### Round 4

Any candidate containing Beer?

Milk-Beer-Bread   
Beer-Butter 

Rejected immediately.

No counting required.

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## Real-World Uses of Apriori

### Amazon

Customers buying:

- Laptop
- Mouse

often buy:

- Laptop Bag

Rule:

Laptop + Mouse  $\rightarrow$  Laptop Bag

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### Grocery Stores

Bread  $\rightarrow$  Butter

or

Diapers → Beer

(the famous market basket example)

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## Netflix

Users watching:

- Money Heist
- Narcos

often watch:

- Breaking Bad
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## Banking

Customers with:

- Salary Account
- Credit Card

often take:

- Personal Loan
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## One-Line Summary

Apriori finds products that frequently occur together by repeatedly building larger item combinations and **pruning any candidate whose smaller subsets are not frequent**, dramatically reducing the search space.